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Section :6

OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

8.To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      foldername  moni      sample1.c
OSLAB                          Simple-Commands-Week1 hello       monis     sample1.c.save
OSLAB14                        example.txt         hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c hello:Zone.Identifier    hello_shortcut skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 2 — Create the Variable Expansion Program

```
Select monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano variable_expansion.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 variable_expansion.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_INPUT 300
#define MAX_OUTPUT 600

void expand_variables(char *input, char *output)
{
    int i = 0;
    int j = 0;

    while (input[i] != '\0' && j < MAX_OUTPUT - 1)
    {
        if (input[i] == '$')
        {
            char variable[100];
            int k = 0;
            char *value;

            i++;

            while (input[i] != '\0' &&
                (isalnum((unsigned char)input[i]) ||
                 input[i] == '_'))
            {
                if (k < 99)
                {
                    variable[k++] = input[i];
                }

                i++;
            }

            variable[k] = '\0';

            value = getenv(variable);

            if (value != NULL)
            {
                int len = strlen(value);

                if (j + len < MAX_OUTPUT - 1)
                {

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^V Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 variable_expansion.c *
    if (value != NULL)
    {
        int len = strlen(value);

        if (j + len < MAX_OUTPUT - 1)
        {
            strcpy(&output[j], value);
            j += len;
        }
    }
    else
    {
        printf("Undefined variable: %s\n", variable);
    }
}
else
{
    output[j++] = input[i++];
}
}
output[j] = '\0';
}

int main()
{
    char input[MAX_INPUT];
    char output[MAX_OUTPUT];

    printf("Enter command: ");

    if (fgets(input, sizeof(input), stdin) == NULL)
    {
        return 1;
    }

    input[strcspn(input, "\n")] = '\0';

    expand_variables(input, output);

    printf("\nExpanded output: [%s]\n", output);

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
```

Save the file

Press:

Ctrl + O

Enter

Ctrl + X

```
Write to File: variable_expansion.c
^G Help      M-D DOS Format  M-A Append      M-B Backup File  ^T Browse
^C Cancel     M-M Mac Format  M-P Prepend     ^O Discard buffer
```

Step 3 — Compile the Variable Expansion Program

Run:

```
gcc -Wall -g variable_expansion.c -o variable_expansion
```

If there are **no errors**, check that the source file and executable exist:

```
ls -l variable_expansion.c variable_expansion
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano variable_expansion.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g variable_expansion.c -o variable_expansion
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l variable_expansion.c variable_expansion
-rwxr-xr-x 1 monis monis 20120 Sep 17 22:09 variable_expansion
-rw-r--r-- 1 monis monis 1579 Sep 17 22:08 variable_expansion.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 4 — Test Variable Expansion

Now run:

`./variable_expansion`

At:

Enter command:

type exactly:

`echo $HOME`

Press Enter.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./variable_expansion
Enter command: echo $HOME

Expanded output: [echo /home/monis]
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 5 — Test an Undefined Variable

Run:

`./variable_expansion`

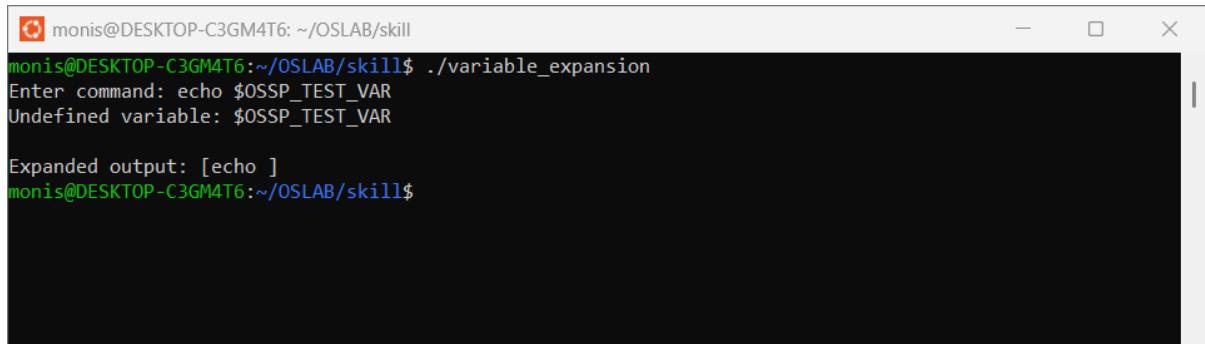
At:

Enter command:

type exactly:

`echo $OSSP_TEST_VAR`

Press Enter.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user enters './variable_expansion'. The terminal shows 'Enter command: echo \$OSSP_TEST_VAR' and 'Undefined variable: \$OSSP_TEST_VAR'. Below that, it shows 'Expanded output: [echo]' and the prompt returns to 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'.

Step 6 — Test Multiple Variable References

Run:

`./variable_expansion`

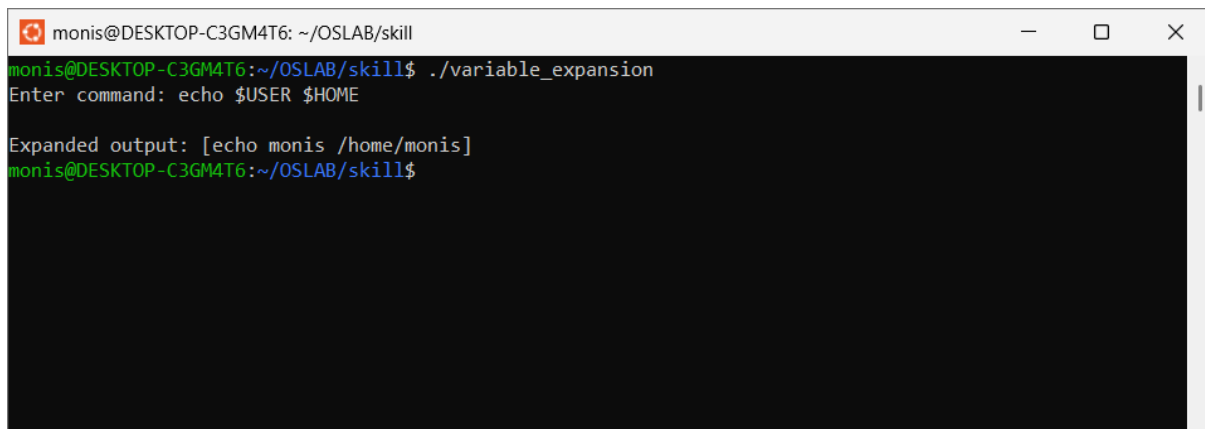
At:

Enter command:

type exactly:

`echo $USER $HOME`

Press Enter.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user enters './variable_expansion'. The terminal shows 'Enter command: echo \$USER \$HOME'. Below that, it shows 'Expanded output: [echo monis /home/monis]' and the prompt returns to 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'.

Step 7 — Test Variable Reference Inside Text

Run:

`./variable_expansion`

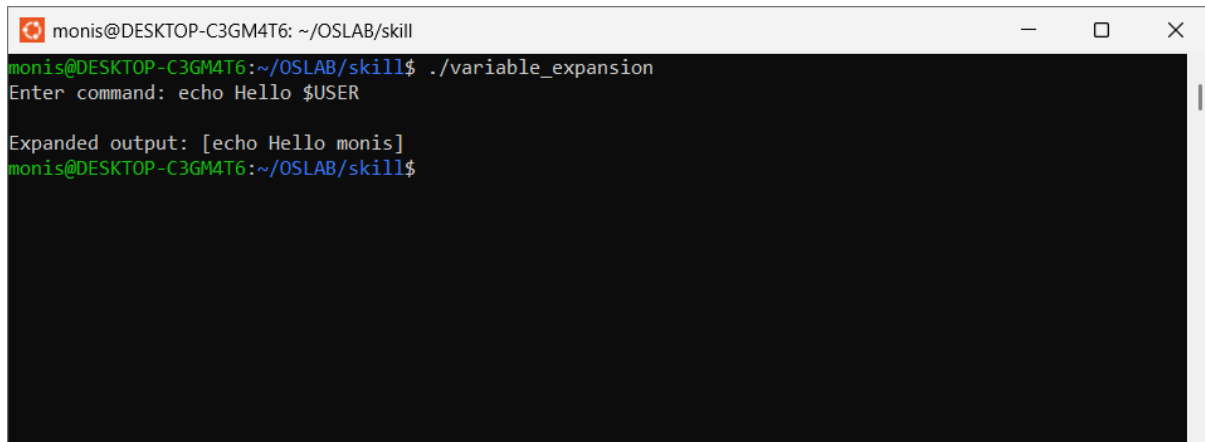
At:

Enter command:

type exactly:

echo Hello \$USER

Press Enter.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user enters './variable_expansion'. The prompt changes to 'Enter command: echo Hello \$USER'. The user presses Enter. The prompt changes to 'Expanded output: [echo Hello monis]'. The user presses Enter again. The prompt returns to 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'.

Step 8 — Test Multiple Variables with Text

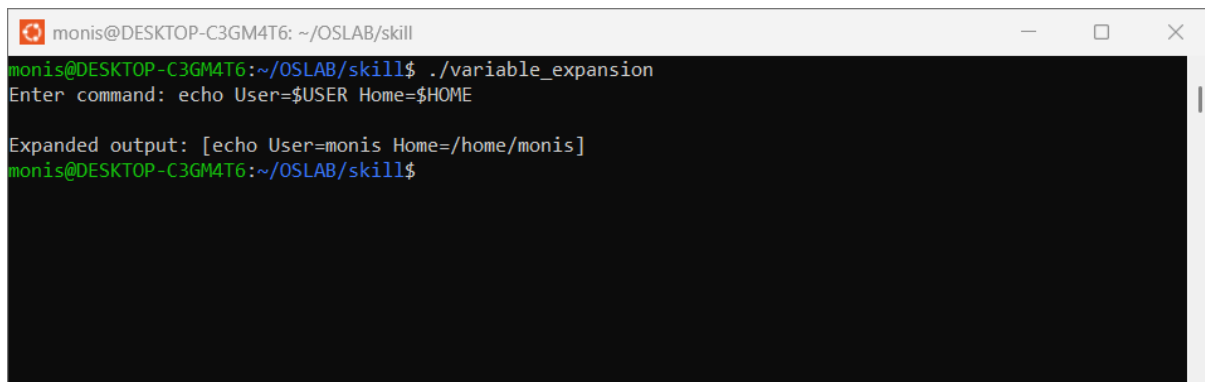
Now run:

./variable_expansion

Enter exactly:

echo User=\$USER Home=\$HOME

Press **Enter**.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user enters './variable_expansion'. The prompt changes to 'Enter command: echo User=\$USER Home=\$HOME'. The user presses Enter. The prompt changes to 'Expanded output: [echo User=monis Home=/home/monis]'. The user presses Enter again. The prompt returns to 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'.

Step 9 — Test Nested Variable References

Run:

./variable_expansion

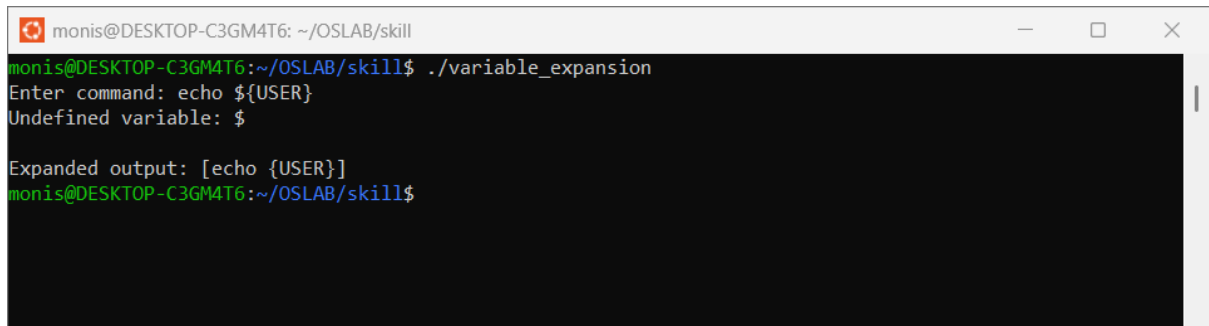
At:

Enter command:

type exactly:

```
echo ${USER}
```

Press Enter.

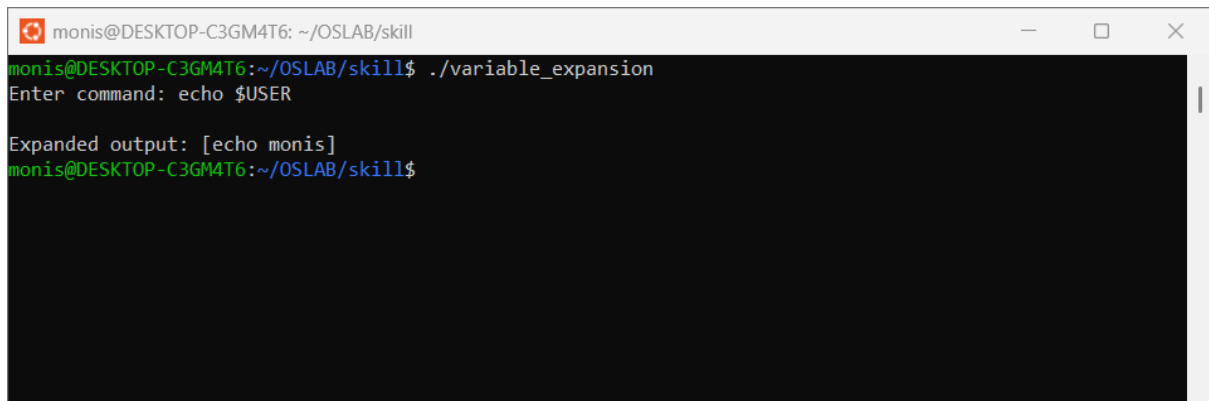
A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user enters './variable_expansion'. The terminal shows 'Enter command: echo \${USER}' and 'Undefined variable: \$'. Then it shows 'Expanded output: [echo {USER}]' and the prompt returns to 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'.

also run:

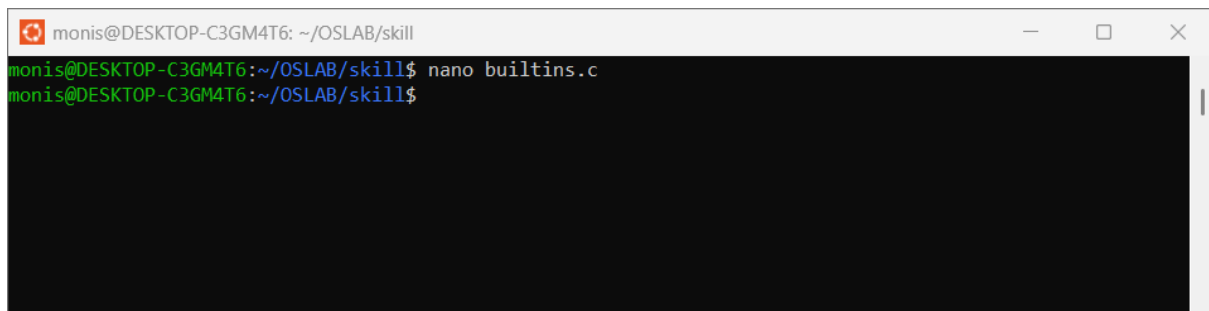
```
./variable_expansion
```

and enter:

```
echo $USER
```

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user enters './variable_expansion'. The terminal shows 'Enter command: echo \$USER'. Then it shows 'Expanded output: [echo monis]' and the prompt returns to 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'.

Step 10 —Built-in Commands

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user enters 'nano builtins.c'. The terminal shows the command and the prompt returns to 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 builtins.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define MAX_INPUT 200
#define MAX_ARGS 20

void builtin_pwd()
{
    char cwd[1024];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
    {
        printf("%s\n", cwd);
    }
    else
    {
        perror("pwd");
    }
}

void builtin_cd(char *path)
{
    if (path == NULL)
    {
        fprintf(stderr, "cd: missing argument\n");
        return;
    }

    if (chdir(path) != 0)
    {
        perror("cd");
    }
}

void builtin_help()
{
    printf("Built-in commands:\n");
    printf("  pwd  - display current directory\n");
    printf("  cd   - change directory\n");
    printf("  help - display built-in commands\n");
    printf("  exit - exit the shell\n");
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1      builtin.c *

}

int main()
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];
    int argc;
    int running = 1;

    printf("Mini Shell Built-ins\n");
    printf("Type help to see available commands.\n");

    while (running)
    {
        printf("\nbuiltin> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
        {
            continue;
        }

        argc = 0;

        char *token = strtok(input, " \t");

        while (token != NULL && argc < MAX_ARGS - 1)
        {
            args[argc++] = token;
            token = strtok(NULL, " \t");
        }

        args[argc] = NULL;

        if (strcmp(args[0], "pwd") == 0)
        {
            builtin_pwd();
        }
    }
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 builtins.c *

    if (strlen(input) == 0)
    {
        continue;
    }

    argc = 0;

    char *token = strtok(input, " \t");

    while (token != NULL && argc < MAX_ARGS - 1)
    {
        args[argc++] = token;
        token = strtok(NULL, " \t");
    }

    args[argc] = NULL;

    if (strcmp(args[0], "pwd") == 0)
    {
        builtin_pwd();
    }
    else if (strcmp(args[0], "cd") == 0)
    {
        builtin_cd(args[1]);
    }
    else if (strcmp(args[0], "help") == 0)
    {
        builtin_help();
    }
    else if (strcmp(args[0], "exit") == 0)
    {
        running = 0;
    }
    else
    {
        printf("Invalid command: %s\n", args[0]);
    }
}

printf("Shell exited.\n");

return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

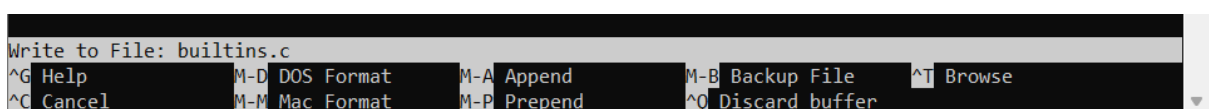
Save

Press:

Ctrl + O

Enter

Ctrl + X



```
Write to File: builtins.c
^G Help    M-D DOS Format    M-A Append    M-B Backup File    ^T Browse
^C Cancel    M-M Mac Format    M-P Prepend    ^Q Discard buffer
```

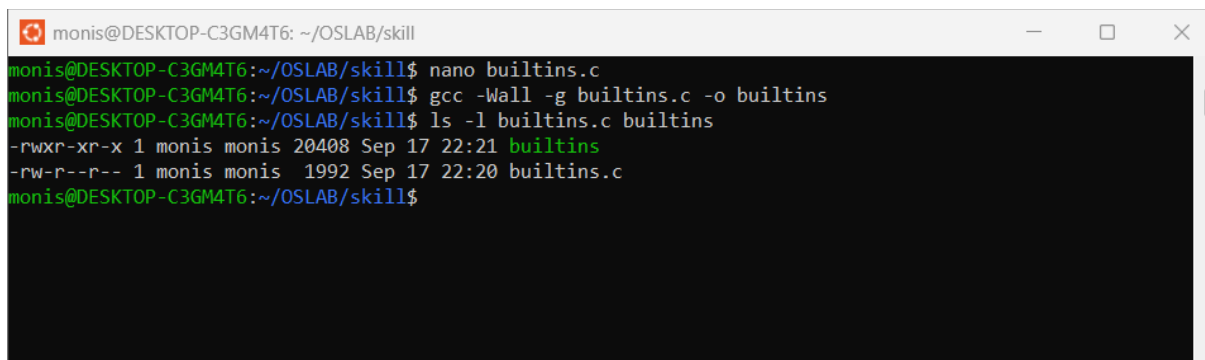
Step 11 — Compile the Built-in Commands Program

Run:

```
gcc -Wall -g builtins.c -o builtins
```

If there are no errors, check that both the source file and executable exist:

```
ls -l builtins.c builtins
```

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill' with standard window controls. The terminal shows the following commands and output:

```
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano builtins.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g builtins.c -o builtins
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l builtins.c builtins
-rwxr-xr-x 1 monis monis 20408 Sep 17 22:21 builtins
-rw-r--r-- 1 monis monis 1992 Sep 17 22:20 builtins.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 12 — Test the help Built-in

Now run:

```
./builtins
```

You should see:

Mini Shell Built-ins

Type help to see available commands.

```
builtin>
```

At builtin>, type:

```
help
```

Press Enter.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtins
Mini Shell Built-ins
Type help to see available commands.

builtin> help
Built-in commands:
  pwd  - display current directory
  cd   - change directory
  help - display built-in commands
  exit - exit the shell

builtin>
```

Step 13 — Test the pwd Built-in

builtin>

Type:

pwd

Press **Enter**.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtins
Mini Shell Built-ins
Type help to see available commands.

builtin> help
Built-in commands:
  pwd  - display current directory
  cd   - change directory
  help - display built-in commands
  exit - exit the shell

builtin> pwd
/home/monis/OSLAB/skill

builtin>
```

Step 14 — Test the cd Built-in

The builtins program should still be running.

First, type:

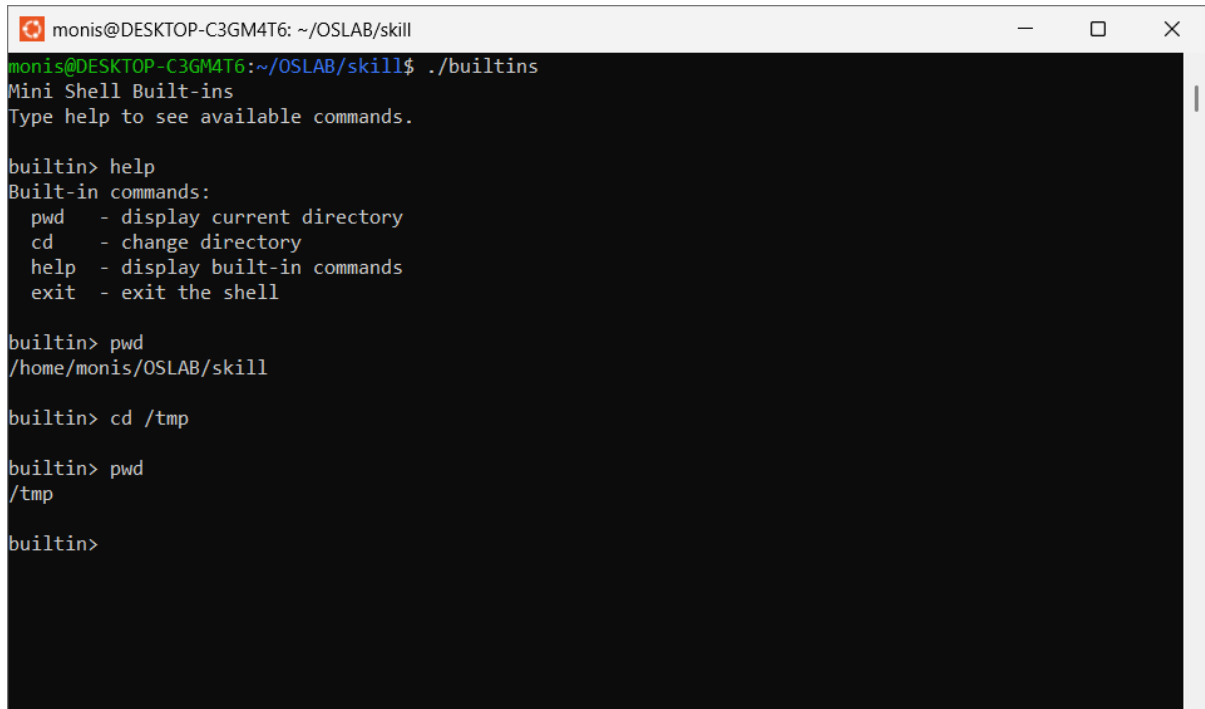
cd /tmp

Press **Enter**.

Then type:

`pwd`

Press **Enter**.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$' and the command './builtins' has been entered. The output shows 'Mini Shell Built-ins' and 'Type help to see available commands.' The prompt changes to 'builtin>'. The user enters 'help', and the output lists built-in commands: 'pwd - display current directory', 'cd - change directory', 'help - display built-in commands', and 'exit - exit the shell'. The user then enters 'pwd', and the output is '/home/monis/OSLAB/skill'. Next, the user enters 'cd /tmp', and the output is '/tmp'. Finally, the user enters 'pwd' again, and the output is '/tmp'. The prompt remains 'builtin>'.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtins
Mini Shell Built-ins
Type help to see available commands.

builtin> help
Built-in commands:
  pwd  - display current directory
  cd   - change directory
  help - display built-in commands
  exit - exit the shell

builtin> pwd
/home/monis/OSLAB/skill

builtin> cd /tmp

builtin> pwd
/tmp

builtin>
```

Step 15 — Test an Invalid Built-in Command

The builtins program should still be running.

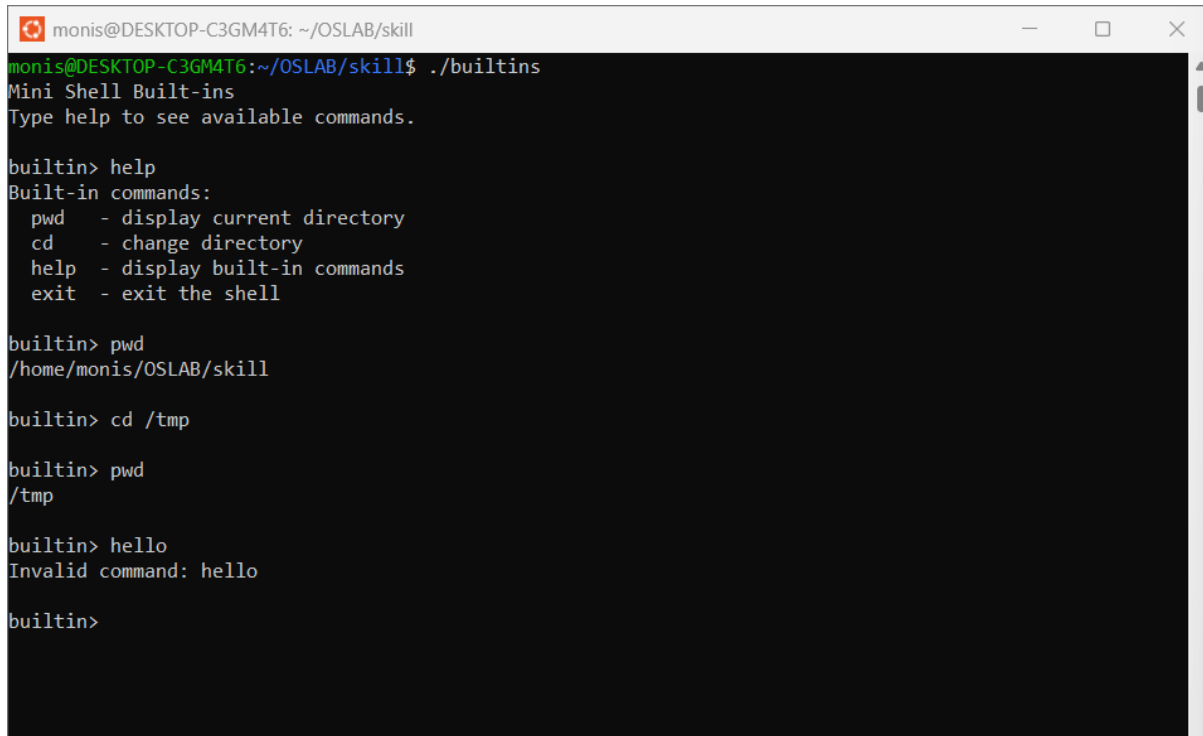
At:

`builtin>`

type:

hello

Press Enter.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtins
Mini Shell Built-ins
Type help to see available commands.

builtin> help
Built-in commands:
  pwd  - display current directory
  cd   - change directory
  help - display built-in commands
  exit - exit the shell

builtin> pwd
/home/monis/OSLAB/skill

builtin> cd /tmp

builtin> pwd
/tmp

builtin> hello
Invalid command: hello

builtin>
```

Step 16 — Test cd Error Handling

The builtins program should still be running.

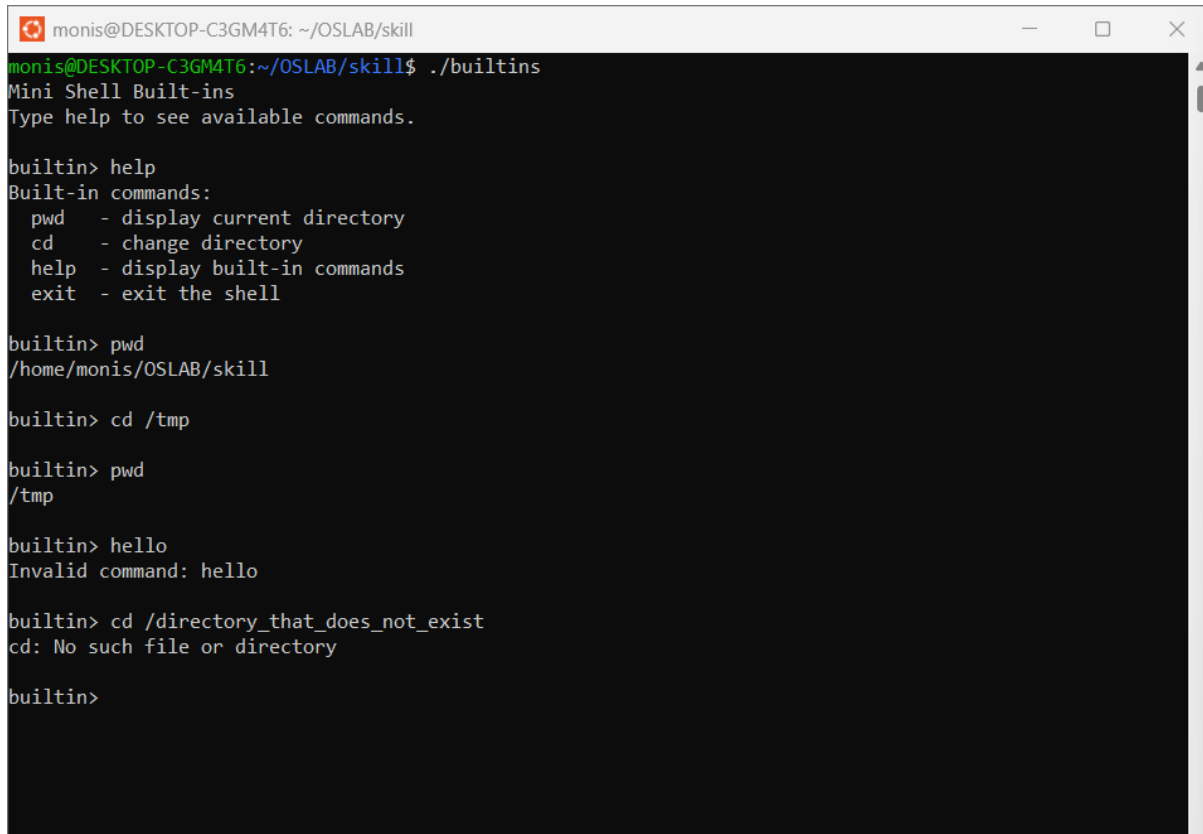
At:

builtin>

type:

cd /directory_that_does_not_exist

Press Enter.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtins
Mini Shell Built-ins
Type help to see available commands.

builtin> help
Built-in commands:
  pwd  - display current directory
  cd   - change directory
  help - display built-in commands
  exit - exit the shell

builtin> pwd
/home/monis/OSLAB/skill

builtin> cd /tmp

builtin> pwd
/tmp

builtin> hello
Invalid command: hello

builtin> cd /directory_that_does_not_exist
cd: No such file or directory

builtin>
```

Step 18 — Test the exit Built-in

The builtins program should still be running.

At:

builtin>

type:

exit

Press **Enter**.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtins
Mini Shell Built-ins
Type help to see available commands.

builtin> help
Built-in commands:
  pwd  - display current directory
  cd   - change directory
  help - display built-in commands
  exit - exit the shell

builtin> pwd
/home/monis/OSLAB/skill

builtin> cd /tmp

builtin> pwd
/tmp

builtin> hello
Invalid command: hello

builtin> cd /directory_that_does_not_exist
cd: No such file or directory

builtin> exit
Shell exited.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```